THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 01

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type **20 questions · 20 marks**

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (0, −7) lies on the:

(a) fourth quadrant (b) third quadrant

(c) x-axis (d) y-axis

Ans: (d) Abscissa is 0, so the point lies on the y-axis.

2. The distance between the points A(−2, 3) and B(4, −5) is:

(a) √28 units (b) 8 units (c) 10 units (d) 14 units

Ans: (c) $d = \sqrt{[(4 + 2)^2 + (-5 - 3)^2]} = \sqrt{(36 + 64)} = 10$.

3. Which of the following is a solution of the linear equation $5x + 2y = 20$?

(a) (2, 5) (b) (3, 2) (c) (5, 2) (d) (4, 1)

Ans: (a) 5(2) + 2(5) = 10 + 10 = 20. The others give 29, 22 and 19.

4. Which of the following is an irrational number?

(a) 0.2525… (25 repeating) (b) √(49/16)

(c) √0.09 (d) √50

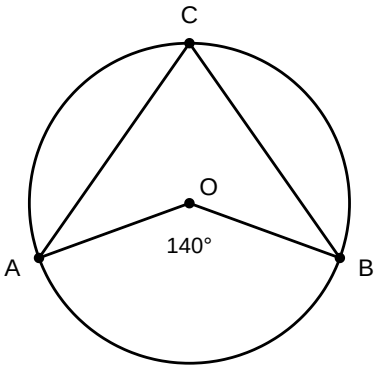
Ans: (d) √50 = 5√2 is irrational. √(49/16) = 7/4, √0.09 = 0.3 and 0.2525… are rational.

5. The product $(x + 4)(x - 4)(x^2 + 16)$ is equal to:

(a) $x^4 - 8x^2 + 256$ (b) $x^4 - 256$ (c) $x^4 + 256$ (d) $x^4 - 16$

Ans: (b) $(x^2 - 16)(x^2 + 16) = x^4 - 256$.

6. In the figure, O is the centre of the circle and ∠AOB = 140°. Then ∠ACB is equal to:



(a) 35° (b) 140° (c) 70° (d) 280°

Ans: (c) Angle at the centre = 2 × angle at the remaining part of the circle, so ∠ACB = 140° ÷ 2 = 70°.

7. A chord of length 48 cm is drawn in a circle of radius 25 cm. Its distance from the centre is:

(a) 24 cm (b) 14 cm (c) 7 cm (d) √1201 cm

Ans: (c) $d^2 = 25^2 - 24^2 = 49$, so d = 7 cm.

8. The sides of a triangle are in the ratio 3 : 5 : 7 and its perimeter is 60 cm. The longest side is:

(a) 12 cm (b) 20 cm (c) 35 cm (d) 28 cm

Ans: (d) One part = 60 ÷ 15 = 4 cm; longest side = 7 × 4 = 28 cm.

9. A letter is chosen at random from the word ‘MATHEMATICS’. The probability that it is a vowel is:

(a) $\frac{4}{11}$ (b) $\frac{5}{11}$ (c) $\frac{7}{11}$ (d) $\frac{1}{4}$

Ans: (a) Vowels A, E, A, I = 4 out of 11 letters.

10. If the n-th term of a sequence is $t_n = 3n^2 - 2$, then t_5 is:

(a) 77 (b) 73 (c) 28 (d) 13

Ans: (b) $t_5 = 3(25) - 2 = 73$.

11. The converse of the proposition ‘If a number is divisible by 15, then it is divisible by 5’ is:

(a) If a number is not divisible by 15, then it is not divisible by 5.
(b) If a number is divisible by 5, then it is divisible by 15.
(c) If a number is divisible by 5, then it is divisible by 3.
(d) A number is divisible by 15 only if it is divisible by 5.

Ans: (b) The converse of ‘if X then Y’ is ‘if Y then X’. (It is false: 10 is a multiple of 5 but not of 15.)

12. A counterexample to the claim ‘If n is a prime number, then $2^n - 1$ is a prime number’ is:

(a) n = 2 (b) n = 7 (c) n = 5 (d) n = 11

Ans: (d) $2^2 - 1 = 3$, $2^5 - 1 = 31$ and $2^7 - 1 = 127$ are prime, but $2^{11} - 1 = 2047 = 23 \times 89$ is composite.

13. In a class, 20 students have an average mark of 60 and the other 30 students have an average mark of 70. The average mark of all 50 students is:

(a) 66 (b) 67 (c) 64 (d) 65

Ans: (a) (20 × 60 + 30 × 70) ÷ 50 = (1200 + 2100) ÷ 50 = 66. (The simple average 65 is wrong because the groups differ in size.)

14. The volume of a cone of base radius 6 cm and height 7 cm (take π = 22/7) is:

(a) 792 cm³ (b) 528 cm³ (c) 264 cm³ (d) 132 cm³

Ans: (c) $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 22/7 \times 36 \times 7 = 264$ cm³ (792 would be $\pi r^2 h$).

15. When the algorithm that checks every j from 1 to n is executed for n = 20, the final list of divisors is:

(a) [1, 2, 4, 5, 10, 15, 20] (b) [2, 4, 5, 10]
(c) [1, 2, 4, 5, 10, 20] (d) [1, 2, 5, 10, 20]

Ans: (c) j = 1, 2, 4, 5, 10 and 20 divide 20; the list stays in increasing order.

16. In a parallelogram ABCD, ∠A = 72°. Then ∠B is equal to:

(a) 118° (b) 108° (c) 98° (d) 72°

Ans: (b) Adjacent angles of a parallelogram are supplementary: ∠B = 180° − 72° = 108°.

17. In △ABC, D and E are the midpoints of AB and AC respectively. If BC = 18 cm, then DE is:

(a) 36 cm (b) 6 cm (c) 9 cm (d) 18 cm

Ans: (c) By the Midpoint Theorem DE = ½ BC = 9 cm.

18. In a sequence $t_1 = 5$ and $t_n = t_{n-1} + 3$ for $n \geq 2$. The sixth term is:

(a) 17 (b) 23 (c) 18 (d) 20

Ans: (d) 5, 8, 11, 14, 17, 20.

19. **Assertion (A):** $\sqrt{3} \times \sqrt{12}$ is a rational number.
Reason (R): The product of two irrational numbers is always irrational.

> **Ans: (c)** $\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$ is rational, so A is true. R is false (e.g. $\sqrt{3} \times \sqrt{12} = 6$, $\sqrt{2} \times \sqrt{2} = 2$).

20. **Assertion (A):** A quadrilateral whose diagonals are equal must be a rectangle.
Reason (R): The diagonals of a rectangle are equal.

> **Ans: (d)** R is true, but its converse is not: an isosceles trapezium has equal diagonals and is not a rectangle, so A is false.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A gym charges ₹800 as joining fee and ₹350 per month. Write an expression for the total amount paid for m months and find the amount for 6 months.

> Total = 800 + 350m (1)
> For m = 6: 800 + 2100 = **₹2900** (1)

22. Without actual division, decide whether $\frac{29}{200}$ has a terminating decimal expansion. If it does, find the number of decimal places.

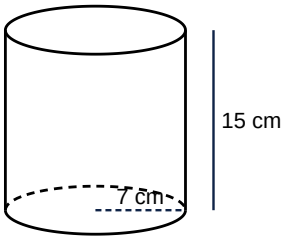
> $200 = 2^3 \times 5^2$, only 2 and 5 occur ⇒ terminates (1). Number of decimal places = larger power = **3** (0.145) (1)

OR

Find two rational numbers between $\frac{1}{6}$ and $\frac{1}{5}$.

> LCM 30 gives 5/30 and 6/30; multiply by 4: 20/120 and 24/120 (1). Two numbers: **21/120 and 22/120** (any two correct numbers) (1)

23. A cylindrical tin of radius 7 cm and height 15 cm is open at the top. Find the area of tin sheet needed to make it ($\pi = 22/7$).

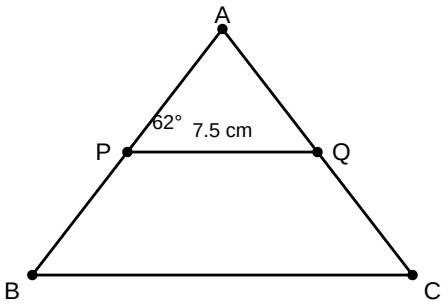


> Open cylinder: area = $2\pi rh + \pi r^2 = \pi r(2h + r)$ (1)
> $= 22/7 \times 7 \times (30 + 7) = 22 \times 37 =$ **814 cm²** (1)

24. Write the converse of the proposition ‘If a number is divisible by 12, then it is divisible by 6.’ Is the converse true? Justify.

> Converse: **If a number is divisible by 6, then it is divisible by 12.** (1)
> False. Counterexample: 18 is divisible by 6 but not by 12. (1)

25. In the figure, P and Q are the midpoints of AB and AC and PQ = 7.5 cm, ∠APQ = 62°. Find BC and ∠ABC, giving reasons.



By the Midpoint Theorem PQ ∥ BC and PQ = ½BC ⇒ **BC = 15 cm** (1)
PQ ∥ BC, so corresponding angles are equal: **∠ABC = ∠APQ = 62°** (1)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.538461538461\ldots = \frac{7}{13}$.

Let x = 0.538461538461… Six digits repeat, so 10^6x = 538461.538461… (1)
Subtracting: 999999x = 538461 (1)
x = 538461/999999 = **7/13** (dividing numerator and denominator by 76923) (1)

27. The mode of travel to school of 100 students was recorded:

Mode	Walk	Cycle	Bus	Car
Students	24	36	28	12

A student is chosen at random. Find the probability that the student (i) cycles, (ii) walks or comes by bus, (iii) does not come by car.

Total = 100. (i) 36/100 = **9/25** (1) (ii) (24 + 28)/100 = **13/25** (1) (iii) 1 − 12/100 = **22/25** (1)

O R

The number of books read last month by 80 students was recorded:

Books	0	1	2	3
Students	12	28	24	16

A student is chosen at random. Find the probability that the student read (i) exactly 2 books, (ii) at least 2 books, (iii) fewer than 2 books.

Total = 80. (i) 24/80 = **3/10** (1) (ii) (24 + 16)/80 = **1/2** (1) (iii) (12 + 28)/80 = **1/2** (1)

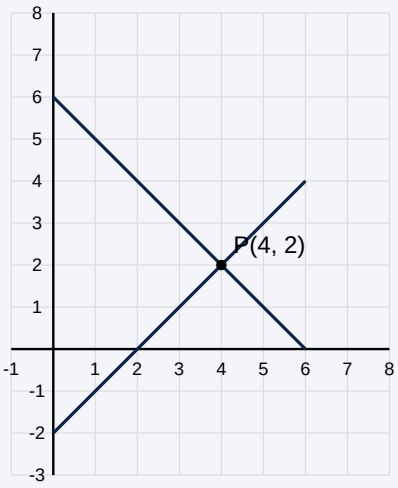
28. Draw the graphs of the equations x + y = 6 and x − y = 2 on the same grid and use them to find the solution of the pair.

Table for x + y = 6:

x	0	6
y	6	0

Table for x − y = 2:

x	0	2
y	-2	0



(1) Draw both lines (1) The lines meet at **(4, 2)**, so x = 4, y = 2 (1)

29. Meera's marks are: unit tests 80%, project 70%, final examination 90%. The annual score is found by combining the three in the ratio 2 : 3 : 5. Find her annual score and say why a simple average of 80, 70 and 90 would not be correct.

Weighted mean = (80 × 2 + 70 × 3 + 90 × 5) ÷ (2 + 3 + 5) (1) = (160 + 210 + 450) ÷ 10 = **82%** (1)
The three components do not carry equal importance; the simple average (80%) ignores their weights. (1)

30. Using Heron's formula, find the area of a triangle whose sides are 9 cm, 10 cm and 17 cm.

s = (9 + 10 + 17)/2 = 18 cm (1)
Area = √[18 × 9 × 8 × 1] (1) = √1296 = **36 cm²** (1)

O R

The sides of a triangle are in the ratio 9 : 12 : 15 and its perimeter is 108 cm. Find its area.

Sides: 27 cm, 36 cm, 45 cm (1); s = 54 (½)
Area = √[54 × 27 × 18 × 9] = √236196 = **486 cm²** (1½)

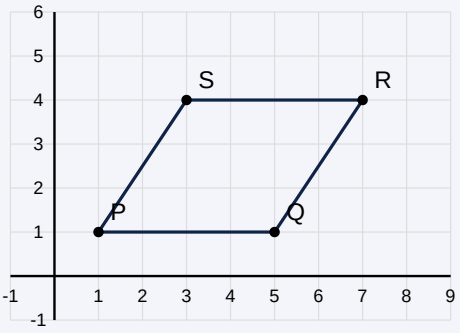
31. Execute the final gcd algorithm of the chapter (start with most-recent-common-divisor = 1 and check every k from 2 to min(m, n)) for m = 9 and n = 12. Show each step and write the gcd.

min(9, 12) = 9; start with 1. (½)
k = 2: does not divide 9 → 1; **k = 3: divides both → 3**; k = 4, 5, 6, 7, 8: not both → stays 3; k = 9: does not divide 12 → stays 3 (2)
gcd(9, 12) = **3** (½)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. A craftsman marks the corners of a metal plate on a grid as P(1, 1), Q(5, 1), R(7, 4) and S(3, 4).
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths PQ, QR, RS and SP. (1½ marks)
(c) Show that PQRS is a parallelogram by finding the midpoints of the diagonals PR and QS. (1½ marks)



(a) Plotting the 4 points (1½) and joining them (½): (b) PQ = 4, RS = 4; QR = √(2² + 3²) = √13, SP = √(2² + 3²) = √13 (1½)
(c) Midpoint of PR = ((1 + 7)/2, (1 + 4)/2) = (4, 2.5); midpoint of QS = ((5 + 3)/2, (1 + 4)/2) = (4, 2.5) (1). The diagonals bisect each other, so PQRS is a parallelogram (½). (Opposite sides equal also accepted.)

33. How many three-digit numbers are multiples of 7? Find their sum. (2½ + 2½)

Numbers: 105, 112, …, 994 — an AP with a = 105, d = 7, l = 994 (1)
n = (994 − 105)/7 + 1 = **128** (1½)
Sum = n/2 (a + l) = 64 × 1099 = **70336** (2½)

O R

A tyre company tested 800 tyres and recorded the distance they ran:

Distance (thousand km)	0 – 20	20 – 40	40 – 60	60 – 80
Number of tyres	40	120	360	280

Find the probability that a tyre chosen at random runs (i) at least 40 thousand km (1), (ii) less than 20 thousand km (1), (iii) between 20 and 60 thousand km (2), and (iv) state the sum of all four class probabilities (1).

Total = 800. (i) (360 + 280)/800 = **4/5** (1) (ii) 40/800 = **1/20** (1) (iii) (120 + 360)/800 = **3/5** (2)
(iv) 40/800 + 120/800 + 360/800 + 280/800 = **1** (1)

34. State and prove the Midpoint Theorem: the segment joining the midpoints of two sides of a triangle is parallel to the third side and equal to half of it.

Statement: in △ABC, if P and Q are midpoints of AB and AC, then PQ ∥ BC and PQ = ½BC. (1)
Proof: Extend PQ to R so that QR = PQ and join CR. (½)
In △APQ and △CRQ: AQ = CQ, PQ = RQ, ∠AQP = ∠CQR (vertically opposite) ⇒ △APQ ≅ △CRQ (SAS) (1½)
So CR = AP = PB and ∠APQ = ∠CRQ ⇒ AB ∥ CR (alternate angles) (1)
PB = CR and PB ∥ CR ⇒ PBCR is a parallelogram (equal parallel sides), so PR ∥ BC and PR = BC. Hence PQ ∥ BC and PQ = ½PR = ½BC. (1)

O R

Batch A of brass weighs 150 kg and is 60% copper. Batch B weighs 100 kg and is 40% copper. Batch C is 45% copper. All three batches are melted together and the alloy is found to be 50% copper. Find the weight of Batch C.

Let Batch C weigh y kg. Copper in A = 90 kg, in B = 40 kg, in C = 0.45y (1½)
Concentration = copper ÷ total: 0.50 = (90 + 40 + 0.45y) ÷ (250 + y) (1½)
0.5(250 + y) = 130 + 0.45y ⇒ 125 + 0.5y = 130 + 0.45y (1)
0.05y = 5 ⇒ **y = 100 kg** (1)

35. The length of a rectangular plot is 4 m more than its breadth and its area is 96 m². Form a quadratic expression in the breadth x, factorise it and find the perimeter of the plot.

Breadth = x m, length = (x + 4) m. Area: x(x + 4) = 96 ⇒ x² + 4x − 96 = 0 (1½)
Factorise: (x + 12)(x − 8) = 0 (1½)

x = 8 (x = −12 rejected) ⇒ breadth 8 m, length 12 m (1)
Perimeter = 2(8 + 12) = **40 m** (1)

OR

A carpenter makes a rectangle from 2 square tiles of side x units, 11 rectangular strips of size x by 1 unit and 12 unit squares. Write the total area as a polynomial, factorise it and find the dimensions of the rectangle.

Area = $2x^2 + 11x + 12$ (1)
Split the middle term: $2x^2 + 8x + 3x + 12$ (1½) = $2x(x + 4) + 3(x + 4) = (2x + 3)(x + 4)$ (1½)
Dimensions: **(2x + 3) units by (x + 4) units** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Ananya saves for a trip. In week 1 she saves ₹150 and every week she saves ₹25 more than in the week before. The amount saved in week n is given by $t_n = 125 + 25n$.

(i) How much does she save in week 12?

$t_{12} = 125 + 25 \times 12 =$ **₹425**

(ii) What type of sequence is this and what is the common difference?

It is an arithmetic progression (each term is 25 more than the previous); common difference = **₹25**.

(iii) In which week does her weekly saving first reach ₹500?

125 + 25n = 500 (1) ⇒ n = 15, i.e. **week 15** (1)

OR

(iii) Find the total amount she saves in the first 10 weeks.

t_{10} = 375 (½). Sum = 10/2 × (150 + 375) (1) = **₹2625** (½)

37. A triangular plot of land has sides 25 m, 39 m and 56 m.

(i) Find the semi-perimeter of the plot.

s = (25 + 39 + 56)/2 = **60 m**

(ii) Find its area using Heron's formula.

Area = √[60 × 35 × 21 × 4] = √176400 = **420 m²**

(iii) Find the length of the perpendicular from the opposite vertex to the longest side.

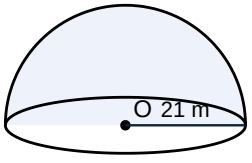
½ × 56 × h = 420 (1) ⇒ h = **15 m** (1)

OR

(iii) The owner fences the plot at ₹80 per metre and levels the ground at ₹12 per m². Find the cost of fencing and the cost of levelling.

Fencing = 120 × 80 = **₹9600** (1); levelling = 420 × 12 = **₹5040** (1)

38. The dome of a planetarium is a hemisphere of radius 21 m (take π = 22/7).



Dome

(i) Find the curved surface area of the dome.

> CSA = $2\pi r^2 = 2 \times 22/7 \times 441 =$ **2772 m²**

(ii) Find the volume of air inside the dome.

> V = $\frac{2}{3}\pi r^3 = \frac{2}{3} \times 22/7 \times 9261 =$ **19404 m³**

(iii) The outside of the dome is painted at ₹5 per m². Find the cost.

> Cost = 2772 × 5 (1) = **₹13860** (1)

O R

(iii) A cylinder has the same radius and height 21 m. Show that volume of dome : volume of cylinder = 2 : 3.

> Cylinder volume = $\pi r^2 h = 22/7 \times 441 \times 21 = 29106$ m³ (1); 19404 : 29106 = **2 : 3** (1)